

SAAS TECHNICAL SPECIFICATION & PITCH DECK

CODENTRAA PORTAL

Enterprise Multi-Tenant Project & Team Management SaaS Platform —
Master Technical Proposal & Developer Blueprint

VERSION

1.0.0 (Production)

PREPARED BY

**Lead Solutions
Architect**

TARGET STACK

**Next.js 14 + ASP.NET
Core 8**

STATUS

**Ready for
Implementation**

100%

Tenant Isolation
Guard

6

Layers

Clean Onion
Architecture

**REST +
WS**

SignalR Realtime
Engine

3 Tiers

SaaS
Monetization
Model

01

Executive Summary & Product Vision

Codentraa is a high-performance, multi-tenant SaaS application designed for organizations, digital agencies, and software development teams to manage workspaces, projects, tasks, file attachments, and real-time team collaboration with complete data privacy.

 Core Value Proposition

Strict logical multi-tenant isolation ensures zero cross-organization data leakage while giving enterprise clients high-speed Next.js SSR interfaces powered by a robust C# ASP.NET Core Web API backend.

Subscription Tier Structure & Monetization Strategy

FEATURE / METRIC	FREE TIER	PRO TIER (\$29/MO)	ENTERPRISE TIER (\$99/MO)
Max Workspaces	1 Organization	3 Organizations	UNLIMITED
Team Members	Up to 5 Users	Up to 50 Users	UNLIMITED
Active Projects	3 Projects	25 Projects	UNLIMITED
Storage Limit	500 MB	50 GB (S3 Cloud)	1 TB DEDICATED
Security & Audit	Basic Logging	Advanced Audit Logs	FULL AUDIT + CUSTOM SLA

02

Role-Based Access Control (RBAC) Permission Matrix

The system enforces strict permission boundaries across 5 organizational roles:

SYSTEM ACTION / CAPABILITY	OWNER	ADMIN	MANAGER	DEVELOPER	CLIENT
Manage SaaS Billing & Subscription	GRANTED	RESTRICTED	RESTRICTED	RESTRICTED	RESTRICTED
Invite / Remove Team Members	GRANTED	GRANTED	RESTRICTED	RESTRICTED	RESTRICTED
Create & Archive Projects	GRANTED	GRANTED	GRANTED	RESTRICTED	RESTRICTED
Create & Assign Tasks	GRANTED	GRANTED	GRANTED	GRANTED	RESTRICTED
Update Task Status & Comment	GRANTED	GRANTED	GRANTED	GRANTED	VIEW & COMMENT ONLY

03 Clean Architecture & High-Level System Design

Codentraa uses a 6-layer Clean Onion Architecture for the backend, enforcing complete separation of concerns and testability.

- | | |
|--|---|
| 1. Presentation Layer | Next.js 14 App Router, React Server Components, Zustand State, Tailwind CSS |
| 2. API Boundary Layer | ASP.NET Core Controllers, Swagger / OpenAPI, CORS Policy, Rate Limiting |
| 3. Application / Business Logic | MediatR CQRS Commands & Queries, FluentValidation, AutoMapper DTOs |
| 4. Multi-Tenant Data Layer | Entity Framework Core 8, TenantDbContextInterceptor (Global Query Filters) |
| 5. Database Layer | SQL Server 2022 (Relational Data, Indexing, Auto-Sequences) |
| 6. Infrastructure Services | ASP.NET Core SignalR (WebSockets), AWS S3 Upload Pre-Signed URLs, SendGrid Mail |

04

Production Database Schema & ERD Specifications

Table: Users

AUTHENTICATION CORE

COLUMN NAME	DATA TYPE	KEY / CONSTRAINT	DESCRIPTION
Id	UNIQUEIDENTIFIER	PRIMARY KEY	Default NEWSEQUENTIALID()
Email	NVARCHAR(256)	UNIQUE, NOT NULL	User login email address
PasswordHash	NVARCHAR(MAX)	NOT NULL	BCrypt / Argon2 Hashed Password
FirstName / LastName	NVARCHAR(100)	NOT NULL	Full user display profile
IsEmailVerified	BIT	DEFAULT 0	Email verification state

Table: Organizations

TENANT ROOT

COLUMN NAME	DATA TYPE	KEY / CONSTRAINT	DESCRIPTION
Id	UNIQUEIDENTIFIER	PRIMARY KEY	Tenant Identifier GUID
Name	NVARCHAR(150)	NOT NULL	Workspace Name (e.g., Acme Corp)
Slug	NVARCHAR(100)	UNIQUE	URL-friendly identifier
OwnerId	UNIQUEIDENTIFIER	FK -> USERS.ID	Organization Owner User GUID

Table: Tasks

PROJECT WORK ITEMS

COLUMN NAME	DATA TYPE	KEY / CONSTRAINT	DESCRIPTION
Id	UNIQUEIDENTIFIER	PRIMARY KEY	Task GUID

COLUMN NAME	DATA TYPE	KEY / CONSTRAINT	DESCRIPTION
OrganizationId	UNIQUEIDENTIFIER	FK (TENANT ISOLATION)	Required for Tenant Filter
ProjectId	UNIQUEIDENTIFIER	FK -> PROJECTS.ID	Parent Project GUID
Title	NVARCHAR(300)	NOT NULL	Task summary title
Status	NVARCHAR(50)	DEFAULT 'Backlog'	Backlog, Todo, InProgress, Review, Done
Priority	NVARCHAR(50)	DEFAULT 'Medium'	Low, Medium, High, Urgent
AssigneeId	UNIQUEIDENTIFIER	FK -> Users.Id (NULL)	Assigned Developer

05 REST API Specification & Endpoint Contracts

HTTP METHOD	ENDPOINT URL	AUTH REQUIRED	DESCRIPTION / ACTION
POST	/api/v1/auth/register	PUBLIC	Register new user account
POST	/api/v1/auth/login	PUBLIC	Returns JWT Access Token & Refresh Cookie
GET	/api/v1/organizations	BEARER JWT	List workspaces owned/joined by user
POST	/api/v1/organizations/{id}/members/invite	BEARER JWT	Send team invite email with role token
GET	/api/v1/projects/{projId}/tasks	BEARER JWT	Get tasks for project (Paginated/Filtered)
PATCH	/api/v1/tasks/{taskId}/status	BEARER JWT	Update task status (Triggers SignalR WebSockets)

06 Senior Developer Action Roadmap & Execution Plan

○ Sprint 1: Solution Infrastructure & Database Foundation

Initialize ASP.NET Core Clean Architecture `.sln`, configure EF Core DbContext, implement `TenantDbContextInterceptor`, and apply SQL migrations.

○ Sprint 2: Authentication & Tenant Resolution Engine

Build JWT Auth Handler, Refresh token rotation, and `TenantResolutionMiddleware` to auto-bind `X-Organization-Id` to requests.

○ Sprint 3: Projects & Task Engine with State Machine Validation

Implement MediatR Commands/Queries for Task CRUD, status state machine validators, and AWS S3 pre-signed upload URL generation.

○ Sprint 4: Next.js 14 Frontend UI & Drag-and-Drop Kanban

Build App Router pages, Sidebar Layout, Interactive Drag-and-Drop Kanban Board with `@hello-pangea/dnd`, and integrate SignalR real-time client.

○ Sprint 5: Executive Dashboard, Stripe Subscriptions & Deployment

Construct metrics charts, integrate Stripe subscription webhooks, create Docker Compose scripts, and deploy to production environment.